Two-span continuous beam, office floor

2026-10-21 · Tragwerk 0.1.0 · EN 1992-1-1:2004 with recommended values · Bikalpa

MAXIMUM UTILISATION

0.8059

Pass

All checks pass
governing: Span 1, bending with axial force

1 Basis of design

ITEM	BASIS
Analysis	Linear-elastic 2D frame analysis (direct stiffness method), Euler–Bernoulli members with axial stiffness. Gross concrete sections, E = E_{cm} (EN 1992-1-1 Table 3.1). Exact internal forces between nodes.
Combinations	EN 1990 Eq. (6.10) for ULS (STR), Eq. 6.14b/6.15b/6.16b for SLS; every variable case on or off (pattern loading); recommended ψ factors (Table A1.1).
Design	EN 1992-1-1:2004 + AC:2010, recommended values of the nationally determined parameters.

PARAMETER	VALUE	CLAUSE
γ_c / γ_s	1.5 / 1.15	2.4.2.4
α_{cc}	1.0	3.1.6(1)
$C_{Rd,c}$ / k_1	0.1200 / 0.15	6.2.2(1)
cot θ range	1.0 – 2.5	6.2.3(2)
ε_{ud}	0.9 · ε_{uk}	3.2.7(2)
Deflection limit	span/250	7.4.1(4)

LOAD FACTOR	VALUE
$\gamma_{G,sup}$ / $\gamma_{G,inf}$	1.35 / 1.0
γ_Q	1.5
ξ (6.10b only)	0.85

2 Structural model



Geometry, supports and member names. Open circles mark moment releases.

MEMBER	NODES	LENGTH [M]	SECTION	DESIGN AS	HINGES
Span 1	A → B	6.000	B300×550	beam	–
Span 2	B → C	6.000	B300×550	beam	–

SUPPORT NODE	RESTRAINED	SPRINGS [KN/M, KNM/RAD]
A	u_x, u_y	–

SUPPORT NODE	RESTRAINED	SPRINGS [KN/M, KNM/RAD]
B	u_y	–
C	u_y	–

SECTION	B × H [MM]	CONCRETE [MPA]	STEEL [MPA]	COVER [MM]	BARS (DEPTH FROM TOP)	LINKS
B300×550	300 × 550	C30/37: f_{cd} = 20.00, f_{ctm} = 2.896, E_{cm} = 32.8 GPa	B500B: f_{yd} = 434.8	30	top: 942 mm² at 48 mm; bottom: 942 mm² at 502 mm	Ø8 × 2 legs @ 200

3 Actions

G – Self weight and finishes (permanent)

- Self weight of all members (25 kN/m³ unless set per concrete)
- Member Span 1: line load -12.00 kN/m, global Y, from 0 to 6.000 m
- Member Span 2: line load -12.00 kN/m, global Y, from 0 to 6.000 m

Q1 – Office load, span 1 (variable, Cat. B, office, ψ_0 = 0.7, ψ_1 = 0.5, ψ_2 = 0.3)

- Member Span 1: line load -10.00 kN/m, global Y, from 0 to 6.000 m

Q2 – Office load, span 2 (variable, Cat. B, office, ψ_0 = 0.7, ψ_1 = 0.5, ψ_2 = 0.3)

- Member Span 2: line load -10.00 kN/m, global Y, from 0 to 6.000 m

4 Combinations of actions

10 ULS, 5 characteristic, 5 frequent and 4 quasi-permanent combinations were generated and all were checked. Combinations that are referenced in this report:

ID	KIND	COMBINATION
ULS004	ULS	1.35·G + 1.50·Q1 + 1.05·Q2
ULS005	ULS	1.35·G + 1.05·Q1 + 1.50·Q2
QP002	SLS qp	1.00·G + 0.30·Q2
QP003	SLS qp	1.00·G + 0.30·Q1

5 Analysis results

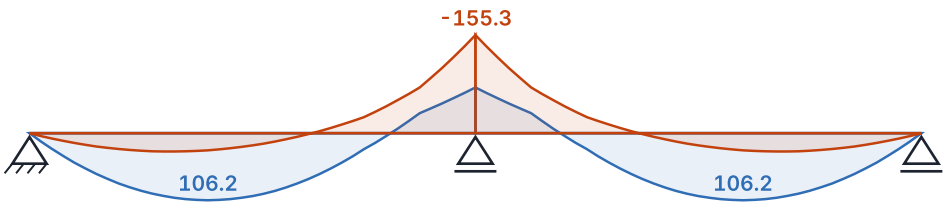
Self-check: for every load case the support reactions balance the applied loads.

LOAD CASE	ΣF_X APPLIED [KN]	ΣF_Y APPLIED [KN]	LARGEST RESIDUAL [KN, KNM]
G	0	-193.5	2.3e-13
Q1	0	-60.00	0.0e+00
Q2	0	-60.00	0.0e+00

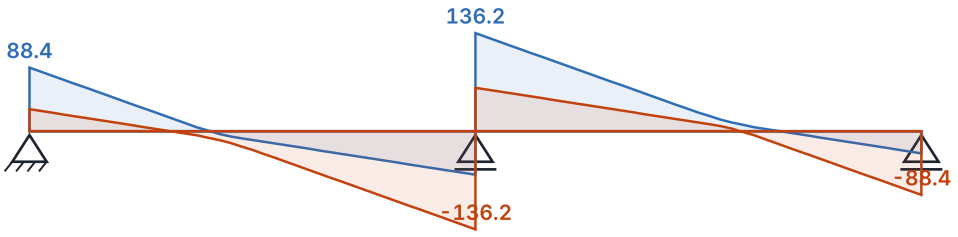
Support reactions

NODE	COMPONENT	G (CHAR.)	Q1 (CHAR.)	Q2 (CHAR.)	ULS MAX	ULS MIN
A	R_y [kN]	36.28	26.25	-3.750	88.35	30.66
B	R_y [kN]	120.9	37.50	37.50	258.9	120.9
C	R_y [kN]	36.28	-3.750	26.25	88.35	30.66

Bending moment, ULS envelope [kNm]
blue: maximum, orange: minimum over all ULS combinations



Shear force, ULS envelope [kN]
blue: maximum, orange: minimum over all ULS combinations



Axial force: zero in every combination.

Deflected shape, CHAR001
max. displacement 0.83 mm, drawn ×1014



Short-term elastic deflections, gross sections, 1.00·G. Long-term deflections with cracking and creep are checked in section 7.

6 Summary of design checks

MEMBER	CHECK	CLAUSE	UTILISATION H	RESULT
Span 1	Bending with axial force	6.1	0.8059	Pass
Span 1	Shear	6.2	0.5517	Pass
Span 1	Maximum longitudinal reinforcement	9.2.1.1(3), 9.5.2(3)	0.2856	Pass
Span 1	Minimum tension reinforcement, bottom	9.2.1.1(1)	0.2407	Pass
Span 1	Minimum tension reinforcement, top	9.2.1.1(1)	0.2407	Pass
Span 1	Minimum shear reinforcement	9.2.2(5)	0.5230	Pass
Span 1	Maximum link spacing	9.2.2(6)	0.5312	Pass
Span 1	Clear bar spacing, bottom	8.2(2)	0.2561	Pass
Span 1	Clear bar spacing, top	8.2(2)	0.2561	Pass
Span 1	Span/depth ratio (deemed to satisfy)	7.4.2	0.5054	Pass
Span 1	Long-term deflection (quasi-permanent)	7.4.3	0.0965	Pass
Span 2	Bending with axial force	6.1	0.8059	Pass
Span 2	Shear	6.2	0.5517	Pass
Span 2	Maximum longitudinal reinforcement	9.2.1.1(3), 9.5.2(3)	0.2856	Pass
Span 2	Minimum tension reinforcement, bottom	9.2.1.1(1)	0.2407	Pass
Span 2	Minimum tension reinforcement, top	9.2.1.1(1)	0.2407	Pass
Span 2	Minimum shear reinforcement	9.2.2(5)	0.5230	Pass
Span 2	Maximum link spacing	9.2.2(6)	0.5312	Pass
Span 2	Clear bar spacing, bottom	8.2(2)	0.2561	Pass
Span 2	Clear bar spacing, top	8.2(2)	0.2561	Pass
Span 2	Span/depth ratio (deemed to satisfy)	7.4.2	0.5054	Pass
Span 2	Long-term deflection (quasi-permanent)	7.4.3	0.0965	Pass

7 Design checks in detail

Each check shows the governing location and combination over all ULS (or quasi-permanent) combinations and all stations along the member, with the formula, the substituted values and the clause.

Member Span 1 · section B300×550 · designed as beam

Bending with axial force · EN 1992-1-1 6.1 · η = 0.8059 — Pass

Governing at x = 6.000 m from node i; combination ULS004: 1.35·G + 1.50·Q1 + 1.05·Q2

f_{cd}	$\alpha_{cc} \cdot f_{ck} / \gamma_c$ = 1.000·30.00/1.500	**20.00** MPa	3.1.6(1)
f_{yd}	f_{yk} / γ_s = 500.0/1.150	**434.8** MPa	3.2.7
ε_{c2}, ε_{cu2}	Table 3.1 = 2.00 ‰, 3.50 ‰	**3.500** ‰	3.1.7
N_{Ed}	axial force (compression +) = 0	**0** kN	
M_{Ed}	bending moment (sagging +) = -155.3	**-155.3** kNm	
x	neutral axis depth from the compressed face (strain compatibility)	**59.79** mm	6.1(2)
$\varepsilon_{c,bottom}$	strain at the compressed face	**3.500** ‰	Fig. 6.1
$\sigma_{s,top}$	$E_s \cdot \varepsilon_s \le f_{yd}$ with ε_s = -25.88 ‰ (bars at depth 48 mm)	**-434.8** MPa	3.2.7
$\sigma_{s,bottom}$	$E_s \cdot \varepsilon_s \le f_{yd}$ with ε_s = 0.69 ‰ (bars at depth 502 mm)	**138.1** MPa	3.2.7
M_{Rd}	resistance moment at N_{Ed} (parabola-rectangle, Fig. 6.1)	**-192.8** kNm	6.1
η	M_{Ed} / M_{Rd} = -155.3/-192.8	**0.8059**	6.1

Shear · EN 1992-1-1 6.2 · η = 0.5517

Pass

Governing at x = 6.000 m from node i; combination ULS004: 1.35·G + 1.50·Q1 + 1.05·Q2

V_{Ed}	design shear force = 136.2	**136.2** kN	
d	effective depth to the top (tension) reinforcement	**502.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/502.0)}$	**1.631**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 942.5/(300.0 \cdot 502.0)$	**0.0063**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 0/165 000	**0** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.631^{1.5} \cdot 30.00^{0.5}$	**0.3994** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.631 \cdot (100 \cdot 0.00626 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0] \cdot 300.0 \cdot 502.0$	**78.35** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·502.0	**451.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 451.8 \cdot 434.8 \cdot 2.500$	**246.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 451.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**493.6** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**246.8** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.2/246.8$	**0.5517**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.2856

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 165\,000$	**6 600.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 885.0** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.2407 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**942.5** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.2407 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**942.5** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5312 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 502.0$	**376.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2561 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 20.00)/2$	**82.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.2561 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 20.00)/2$	**82.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.5054

Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0063**	
ρ'	compression reinforcement ratio	**0**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/(\rho - \rho') + (1/12)\cdot\sqrt{f_{ck}}\cdot\sqrt{(\rho'/\rho_0)}]$, ρ' taken as 0 (conservative; the equation is singular as ρ' → ρ)	**23.65**	Eq. 7.16b
l/d	span/effective depth = 6000.0/502.0	**11.95**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.0965

Pass

Governing at x = 2.550 m from node i; combination QP003: 1.00·G + 0.30·Q1

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**5 837.0** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**2 627.4** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**61.48** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0**	Eq. 7.19
w_{max}	$\int \kappa\cdot\bar{m}\,dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.317** mm	Eq. 7.18
w_{lim}	span/250 = 6000.0/250	**24.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.317/24.00	**0.0965**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

Member Span 2 · section B300×550 · designed as beam

Bending with axial force · EN 1992-1-1 6.1 · η = 0.8059 — Pass

Governing at x = 0.000 m from node i; combination ULS004: 1.35·G + 1.50·Q1 + 1.05·Q2

f_{cd}	$\alpha_{cc} \cdot f_{ck} / \gamma_c$ = 1.000·30.00/1.500	**20.00** MPa	3.1.6(1)
f_{yd}	f_{yk} / γ_s = 500.0/1.150	**434.8** MPa	3.2.7
ε_{c2}, ε_{cu2}	Table 3.1 = 2.00 ‰, 3.50 ‰	**3.500** ‰	3.1.7
N_{Ed}	axial force (compression +) = 0	**0** kN	
M_{Ed}	bending moment (sagging +) = -155.3	**-155.3** kNm	
x	neutral axis depth from the compressed face (strain compatibility)	**59.79** mm	6.1(2)
$\varepsilon_{c,bottom}$	strain at the compressed face	**3.500** ‰	Fig. 6.1
$\sigma_{s,top}$	$E_s \cdot \varepsilon_s \le f_{yd}$ with ε_s = -25.88 ‰ (bars at depth 48 mm)	**-434.8** MPa	3.2.7
$\sigma_{s,bottom}$	$E_s \cdot \varepsilon_s \le f_{yd}$ with ε_s = 0.69 ‰ (bars at depth 502 mm)	**138.1** MPa	3.2.7
M_{Rd}	resistance moment at N_{Ed} (parabola-rectangle, Fig. 6.1)	**-192.8** kNm	6.1
η	M_{Ed} / M_{Rd} = -155.3/-192.8	**0.8059**	6.1

Shear · EN 1992-1-1 6.2 · η = 0.5517

Pass

Governing at x = 0.000 m from node i; combination ULS005: 1.35·G + 1.05·Q1 + 1.50·Q2

V_{Ed}	design shear force = 136.2	**136.2** kN	
d	effective depth to the top (tension) reinforcement	**502.0** mm	
k	$1 + \sqrt{(200/d)} \le 2.0$ $= 1 + \sqrt{(200/502.0)}$	**1.631**	6.2.2(1)
ρ_l	$A_{sl}/(b_w \cdot d) \le 0.02$ $= 942.5/(300.0 \cdot 502.0)$	**0.0063**	6.2.2(1)
σ_{cp}	$N_{Ed}/A_c < 0.2 \cdot f_{cd}$ = 0/165 000	**0** MPa	6.2.2(1)
v_{min}	$0.035 \cdot k^{3/2} \cdot f_{ck}^{1/2}$ $= 0.035 \cdot 1.631^{1.5} \cdot 30.00^{0.5}$	**0.3994** MPa	6.2.2(1) Eq. 6.3N
$V_{Rd,c}$	$[C_{Rd,c} \cdot k \cdot (100 \cdot \rho_l \cdot f_{ck})^{1/3} + k_1 \cdot \sigma_{cp}] \cdot b_w \cdot d \ge (v_{min} + k_1 \cdot \sigma_{cp}) \cdot b_w \cdot d$ $= [0.12 \cdot 1.631 \cdot (100 \cdot 0.00626 \cdot 30.00)^{1/3} +$ $0.15 \cdot 0] \cdot 300.0 \cdot 502.0$	**78.35** kN	6.2.2(1) Eq. 6.2
A_{sw}/s	2 legs Ø8 at 200 mm = 100.5/200.0	**0.5027** mm²/mm	
z	0.9·d = 0.9·502.0	**451.8** mm	6.2.3(1)
cot θ	chosen in [1, 2.5] to maximise $\min(V_{Rd,s}, V_{Rd,max})$	**2.500**	6.2.3(2) Eq. 6.7N
$V_{Rd,s}$	$(A_{sw}/s) \cdot z \cdot f_{ywd} \cdot \cot\theta$ $= 0.503 \cdot 451.8 \cdot 434.8 \cdot 2.500$	**246.8** kN	6.2.3(3) Eq. 6.8
$V_{Rd,max}$	$\alpha_{cw} \cdot b_w \cdot z \cdot v_1 \cdot f_{cd}/(\cot\theta + \tan\theta)$ $= 1.0 \cdot 300.0 \cdot 451.8 \cdot 0.528 \cdot 20.00/(2.500 + 0.4)$	**493.6** kN	6.2.3(3) Eq. 6.9
V_{Rd}	$\max(V_{Rd,c}, \min(V_{Rd,s}, V_{Rd,max}))$	**246.8** kN	6.2.1
η	V_{Ed}/V_{Rd} $= 136.2/246.8$	**0.5517**	6.2.1

- Checked at every station, including at the support axis (conservative; 6.2.1(8) allows d from the face).

Maximum longitudinal reinforcement · EN 1992-1-1 9.2.1.1(3), 9.5.2(3) · η = 0.2856

Pass

$A_{s,max}$	$0.04 \cdot A_c$ $= 0.04 \cdot 165\,000$	**6 600.0** mm²
$A_{s,prov}$	all longitudinal bars	**1 885.0** mm²

- Limit applies outside lap locations.

Minimum tension reinforcement, bottom · EN 1992-1-1 9.2.1.1(1) · η = 0.2407 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	bottom bars	**942.5** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum tension reinforcement, top · EN 1992-1-1 9.2.1.1(1) · η = 0.2407 — Pass

f_{ctm}	Table 3.1	**2.896** MPa	3.1.2
$A_{s,min}$	$\max(0.26 \cdot f_{ctm}/f_{yk}, 0.0013) \cdot b_t \cdot d$ $= \max(0.26 \cdot 2.896/500.0, 0.0013) \cdot 300.0 \cdot 502.0$	**226.8** mm²	Eq. 9.1N
$A_{s,prov}$	top bars	**942.5** mm²	

- Ratio shown is $A_{s,min}/A_{s,prov}$.

Minimum shear reinforcement · EN 1992-1-1 9.2.2(5) · η = 0.5230 — Pass

ρ_w	$A_{sw}/(s \cdot b_w)$ $= 100.5/(200.0 \cdot 300.0)$	**0.0017**	Eq. 9.4
$\rho_{w,min}$	$0.08 \cdot \sqrt{f_{ck}}/f_{yk}$ $= 0.08 \cdot \sqrt{30.00}/500.0$	**8.764e-04**	Eq. 9.5N

Maximum link spacing · EN 1992-1-1 9.2.2(6) · η = 0.5312 — Pass

$s_{l,max}$	$0.75 \cdot d \cdot (1 + \cot \alpha)$, $\alpha = 90°$ $= 0.75 \cdot 502.0$	**376.5** mm	Eq. 9.6N

Clear bar spacing, bottom · EN 1992-1-1 8.2(2) · η = 0.2561 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 20.00)/2$	**82.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Clear bar spacing, top · EN 1992-1-1 8.2(2) · η = 0.2561 — Pass

a	$(b - 2 \cdot c - 2 \cdot \varphi_w - n \cdot \varphi)/(n - 1)$ $= (300.0 - 2 \cdot 30.00 - 2 \cdot 8.000 - 3 \cdot 20.00)/2$	**82.00** mm
a_{min}	$\max(k_1 \cdot \varphi, d_g + k_2, 20\ mm)$ $= \max(20.00, 16.00 + 5, 20)$	**21.00** mm

- Assumes one layer of bars across the width.

Span/depth ratio (deemed to satisfy) · EN 1992-1-1 7.4.2 · η = 0.5054 — Pass

K	structural system: end span	**1.300**	Table 7.4N
ρ_0	$\sqrt{f_{ck}}\cdot 10^{-3}$ $= \sqrt{30.00}\cdot 10^{-3}$	**0.0055**	7.4.2(2)
ρ	$A_s/(b\cdot d)$, bottom bars as provided (conservative)	**0.0063**	
ρ'	compression reinforcement ratio	**0**	
$(l/d)_{lim}$	$K\cdot[11 + 1.5\cdot\sqrt{f_{ck}}\cdot\rho_0/(\rho - \rho') + (1/12)\cdot\sqrt{f_{ck}}\cdot\sqrt{(\rho'/\rho_0)}]$, ρ' taken as 0 (conservative; the equation is singular as ρ' → ρ)	**23.65**	Eq. 7.16b
l/d	span/effective depth = 6000.0/502.0	**11.95**	

- Uses the provided reinforcement and omits the 310/σ_s factor, both conservative.

Long-term deflection (quasi-permanent) · EN 1992-1-1 7.4.3 · η = 0.0965 — Pass

Governing at x = 3.450 m from node i; combination QP002: 1.00·G + 0.30·Q2

$E_{c,eff}$	$E_{cm}/(1 + \varphi(\infty,t_0))$ $= 32836.6/(1 + 2.000)$	**10 945.5** MPa	Eq. 7.20
α_e	$E_s/E_{c,eff}$	**18.27**	
I_I	uncracked transformed section, sagging	**5 837.0** $\times 10^6$ mm⁴	
I_{II}	fully cracked transformed section, sagging	**2 627.4** $\times 10^6$ mm⁴	
M_{cr}	$f_{ctm}\cdot I_I/(h - y_I)$, sagging	**61.48** kNm	7.4.3(3)
ζ	$1 - \beta\cdot(M_{cr}/M_{qp})^2$, β = 0.5, at the governing station	**0**	Eq. 7.19
w_{max}	$\int \kappa\cdot\bar{m}\,dx$ with $\kappa = \zeta\cdot\kappa_{II} + (1 - \zeta)\cdot\kappa_I$ (relative to the member chord)	**2.317** mm	Eq. 7.18
w_{lim}	span/250 = 6000.0/250	**24.00** mm	7.4.1(4)
η	w_{max}/w_{lim} = 2.317/24.00	**0.0965**	

- Moments from the linear-elastic analysis (no redistribution from cracking).
- Shrinkage curvature (7.4.3(6)) is not included; add a margin or check it separately.

8 Assumptions and limitations

- Plane frame; out-of-plane behaviour, torsion and lateral-torsional effects are not considered.
- Linear-elastic analysis with gross sections; no moment redistribution, no second-order (P-Δ) effects and no geometric imperfections (5.2).
- Reinforcement is assumed uniform along each member (no curtailment); anchorage and laps (8.4–8.7) are not checked.
- Shear is checked at every station including the support axis; torsion and punching are not covered.
- Columns: slender columns ($\lambda > \lambda_{lim}$) are flagged as not covered; biaxial bending is not considered.
- Deflection: curvature interpolation (7.4.3) using elastic moments; shrinkage curvature is excluded. Crack widths (7.3) and stress limits (7.2) are not yet checked.
- Durability, cover requirements (4.4) and fire resistance are the responsibility of the designer.
- A licensed engineer remains responsible for the model, its inputs and the use of these results.

9 Software verification

This build of Tragwerk 0.1.0 ran its verification suite when producing this report: 25 benchmarks with 114 comparisons against closed-form solutions and hand calculations, **all passed**. The solver is also cross-checked against two independent programs in the test suite.

Generated by Tragwerk 0.1.0 on 2026-10-21. Results are reproducible from the model file; analysis time 0.36 s.